Brian Pigott, Ph.D.

Contact Information

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Experience

Current June 2021	Associate Professor and Chair *Department of Mathematics, Wofford College* *Chair as of Fall 2025*
September 2021 –September 2023	Advisory Board Member *Rosalyn.ai*
February 2022 – August 2022	QuForce Innovation Fellow
September 2014 – June 2021	Assistant Professor *Department of Mathematics, Wofford College*
July 2011 – July 2014	Teacher-Scholar Postdoctoral Fellow *Department of Mathematics, Wake Forest University*

Education

2011 **University of Toronto**, Ph.D. in Mathematics
Dissertation: *Low Regularity Stability for Subcritical Generalized Korteweg-de Vries Equations*
Advisor: James Colliander

2002 **University of Toronto**, M.Sc. in Mathematics
Research Project: Nuts and Bolts of Cyclic Homology
Advisor: George Elliott

2001 **University of Calgary**, B.Sc. (Honours) in Mathematics

Research Interests

Mathematical Analysis, Partial Differential Equations, Quantum Computing and Quantum Information, Quantum Dynamics

Publications

Published

2025 B. Pigott, E. Campolongo, H. Routray, and A. Khan, *Eavesdropping on the BB84 Protocol using Phase-Covariant Cloning: Experimental Results*, International Journal of Quantum Information, 23(7) (2025)

2023 N. Kosovalic and B. Pigott, *Hopf-like bifurcation in a wave equation at a removable singularity*, Journal of Differential Equations, **369** (383-404)

2022 N. Kosovalic and B. Pigott, *Symmetric vibrations of higher dimensional nonlinear wave equations*, Selecta Mathematica (N.S.), 28(3), Paper No. 48

2020 N. Kosovalic and B. Pigott, *Uniqueness for sums of nonvanishing squares*, Integers, **20**, Paper No. A93.

2019 N. Kosovalic and B. Pigott, *Self-excited vibrations for damped and delayed 1-dimensional wave equations*, Journal of Dynamics and Differential Equations, **31**, no. 1 (129-152)

2019 N. Kosovalic and B. Pigott, *Self-excited vibrations for damped and delayed higher dimensional wave equations*, Discrete and Continuous Dynamical Systems A, **39**, no. 5 (2413-2435).

2018 L.G. Farah and B. Pigott, *Nonlinear profile decomposition and the concentration phenomenon for supercritical generalized KdV equations*, Indiana Univ. Math. Journal, **67**, no. 5 (1857–1892)

2017 B. Pigott and S. Raynor, *Long-term stability for KdV solitons in weighted H^s spaces*, Commun. Pure Appl. Anal., **16**, no.2 (393-416).

2016 B. Pigott, *On mass concentration for the critical generalized KdV equation*, Proc. Edinburgh Math. Soc., **59**, no. 2 (519-532).

2014 B. Pigott and S. Raynor, *Asymptotic stability for KdV solitons in weighted spaces via iteration*, Illinois J. Math., **58**, no. 2 (443-470).

2014 B. Pigott, *Polynomial-in-time upper bounds for the orbital instability of subcritical generalized Korteweg-de Vries equations*, Commun. Pure Appl. Anal., **13**, no. 1, (389-418).

2003 J. Bryden, T. Lawson, B. Pigott, and P. Zvengrowski, *The integral homology of orientable Seifert manifolds*, Topology Appl., **127**, no. 1-2 (259-275).

In Preparation

2026 J. Kraisler and B. Pigott, *Effective Dynamics in Many Photon Real Space Quantum Optics I*

2026 J. Kraisler and B. Pigott, *Effective Dynamics in Many Photon Real Space Quantum Optics II*

Undergraduate Research Projects Supervised

Spring 2025	L. Frey, *Bifurcation Theory of Burgers Equation*.
Spring 2025	W. Theo, *Introduction to Quantum Computing*.
2024-2025	J. Daughtry, *Distribution of Entanglement Under Universal Cloning Transformations*.
2018-2019	H. Wages, *Random Walks and the Relativistic Heat Equation*.
2016-2017	D. Couch, *Quantum Graphs and a Model of the Nervous System*.
Interim 2016, Spring 2016	J. Cohen, *A Statistical Analysis of* Big Brother
Fall 2015	D. Tuyisenge, *Inverse Scattering for the Korteweg-de Vries Equation*.
Summer 2013	R. Dougherty, *The Multilinear Correction Algorithm and Conservation Laws for the Korteweg-de Vries Equation*, jointly supervised with F. Moore.
Summer 2013	J. Byrum, *The Inverse Scattering Transform and Scale Invariant Lax Pairs*.

Teaching Experience

Instructor

2014- **Wofford College**

Introduction to Statistics, Preparation for Calculus: Functions Modeling Change, Calculus I, Calculus II, Multivariable Calculus, Vector Calculus, Discrete Mathematical Models, Ordinary Differential Equations, Partial Differential Equations, Research: Inverse Scattering for the Korteweg-de Vries Equation, Introduction to Calculus Pedagogy, Functions of a Complex Variable, Topology, Introduction to Quantum Computing, Bifurcation Theory of Burgers Equation

2011-2014 **Wake Forest University**

Calculus with Analytic Geometry I, Multivariable Calculus, Advanced Calculus, Ordinary Differential Equations, Partial Differential Equations, Reading in Mathematics: Harmonic Analysis, Real Analysis: Measure Theory, Topics in Analysis: C^*-Algebras

Course Coordinator

2010-2011 MAT 137 Calculus!, University of Toronto, Faculty of Arts and Science.

Coordinated all aspects of a course with over 500 students in 6 lecture sections and 20 tutorials, including weekly problem sets, three term tests, four quizzes, and a final exam.

2007-2011 **University of Toronto**

Calculus A, Calculus B, Calculus I, Calculus II, Advanced Engineering Mathematics

2009 **Georgian College**

Mathematics for Technology, Introduction to Technical Mathematics

Selected Seminars and Talks

April 2025 Department of Mathematics Colloquium, Wake Forest University, Winston-Salem NC, USA

November 2019 Department of Mathematics and Statistics Colloquium, University of South Alabama, Mobile AL, USA

November 2019 American Mathematical Society Fall Southeastern Sectional Meeting, University of Florida, Gainesville FL, USA, *Special Session on Analysis of Geometric and Evolutionary PDEs*

April 2015 Student Physics Seminar, Wofford College, Spartanburg SC, USA

March 2015 American Mathematical Society Spring Southeastern Sectional Meeting, University of Alabama at Huntsville, Huntsville AL, USA. *Special Session on Recent Progress in Differential Equations*

November 2013 Applied Math Seminar, University of North Carolina at Greensboro, Greensboro NC, USA.

November 2013 Differential Equations Seminar, North Carolina State University, Raleigh NC, USA

November 2013 Applied Math Club Seminar, North Carolina State University, Raleigh NC, USA

October 2013 American Mathematical Society Fall Eastern Sectional Meeting, Temple University, Philadelphia PA, USA. *Special Session on Nonlinear Elliptic and Wave Equations and Applications*

October 2012 Southeastern-Atlantic Regional Conference on Differential Equations (SEARCDE), Wake Forest University, Winston-Salem NC, USA

October 2012 Wake Forest Math Club Talk, Wake Forest University, Winston-Salem NC, USA

September 2011	American Mathematical Society Fall Southeastern Sectional Meeting, Wake Forest University, Winston-Salem NC, USA. *Special Session on Nonlinear Dispersive Equations*
February 2011	Wake Forest University Colloquium, Wake Forest University, Winston-Salem NC, USA

Conferences and Workshops

Conferences Organized

May 2027	SIAM Conference on Applications of Dynamical Systems, Atlanta GA, USA. *Effective Models and Approximate Dynamics in Quantum Systems* (joint with J. Kraisler)
March 2017	American Mathematical Society Spring Southeastern Sectional Meeting, College of Charleston, Charleston SC, USA. *Special Session on Advances in Long-Term Behavior of Nonlinear Dispersive Equations* (joint with S. Raynor)
October 2012	Southeastern-Atlantic Regional Conference on Differential Equations (SEARCDE), Wake Forest University, Winston-Salem NC, USA. (joint with S. Raynor and S. Robinson)

Conferences Attended

February 2025	Quantum Information Processing, Raleigh NC, USA
October 2024	SIAM Quantum Intersections Convening, Tysons Corner VA, USA
February 2023	QHack, sponsored by Xanadu (remote)
January 2023	qBraid Conference, sponsored by qBraid (remote)
February 2022	QHack, sponsored by Xanadu (remote)
February 2019	International Association for Mathematics and Computers in Simulation, *Conference on Nonlinear Evolution equations and Wave Phenomena: Computation and Theory*, Athens GA, USA
June 2017	French-American Conference on Nonlinear Dispersive PDEs, Centre International de Rencontres Mathématiques, Marseilles, FR.
May 2016	John H. Barrett Memorial Lectures, University of Tennessee, Knoxville, Knoxville TN, USA
January 2014	Joint Mathematics Meetings, Baltimore MD, USA
June 2013	NSF-CBMA Regional Research Conference in the Mathematical Sciences, *The Global Behavior of Solutions to Critical Nonlinear Wave Equations*, Kansas State University, Manhattan KS, USA
March 2013	International Association for Mathematics and Computers in Simulation, *Conference on Nonlinear Evolution equations and Wave Phenomena: Computation and Theory*, Athens GA, USA
January 2013	Joint Mathematics Meetings, San Diego CA, USA
March 2012	American Mathematical Society Spring Eastern Sectional Meeting, George Washington University, Washington DC, USA
April 2006	Schrödinger Evolution Equations, Banff International Research Station, Banff AB, Canada
June 2004	Canadian Operator Symposium, University of Waterloo, Waterloo ON, Canada
December 2002	Canadian Mathematical Society Winter Meeting, Ottawa ON, Canada
May 2001	Cascade Topology Seminar, Boise State University, Boise ID, USA
July 1999	Invariants of 3-Manifolds, Morley AB, Canada.

Service

Ongoing	Referee for various professional mathematics journals
Ongoing	Reviewer for Math Reviews
Fall 2025 - Present	Chair, Department of Mathematics, Wofford College
Spring 2023 - Present	Faculty Marshal, Wofford College Opening Convocation and Commencement Weekend
February 2026	Co-Organizer for Joint Math-Physics Colloquium: "Implantable Brain-Computer Interfaces: Toward the Restoration of Communication and Mobility", by Dr. Leigh Hochberg, Brown University and Harvard Medical School, Wofford College (joint with C. Martsberger)
February 2025	Co-Organizer for Joint Math-Physics Colloquium: "Quantum Technology: Past, Present, and Future", by Dr. Jungsang Kim, Duke Quantum Center, Wofford College (joint with C. Martsberger)
Spring 2025	External Evaluator for Promotion to Professor, Teaching Stream, University of Toronto Mississauga
2024-2025	Faculty Handbook Revision Committee, Wofford College
Spring 2024	Department Chair (Sabbatical Replacement), Department of Mathematics, Wofford College
Spring 2024	Piedmont Region III Science Fair Judge, Spartanburg SC, USA
Spring 2024	Search Committee, Department of Mathematics, Wofford College
2023 - 2024	Search Committee, Department of Philosophy, Wofford College
2022 - 2023	Search Committee, Department of Computer Science, Wofford College
Fall 2019 - present	Benefits Committee, Wofford College
Fall 2019	Search Committee, Department of Mathematics, Wofford College.
Fall 2017 - present	Department of Mathematics Colloquium Organizer, Wofford College
2018 - 2019	Calculus Task Force, Department of Mathematics, Wofford College (Chair)
2015 - 2018	Member of the Library Committee, Wofford College (Chair from Fall 2017-Spring 2018)
2014 - 2017	Department of Mathematics Colloquium Organizer, Wofford College (joint with J. Spivey)
February 2017	Piedmont Region III Science Fair Judge, Spartanburg SC, USA
February 2016	Piedmont Region III Science Fair Judge, Spartanburg SC, USA
February 2015	Piedmont Region III Science Fair Judge, Spartanburg SC, USA
Spring 2015	Faculty Mentor for the Collaborative Research Program in Mathematics, Wofford College
2012 - 2014	Member of the Analysis Committee, Department of Mathematics, Wake Forest University
2012 - 2014	Member of the Graduate Committee, Department of Mathematics, Wake Forest University
2011 - 2014	Member of the Math Club Committee, Department of Mathematics, Wake Forest University
2012 - 2013	Member of the Wake Forest University Postdoctoral Association Board, Wake Forest University
2002 - 2005	Mentor for the University of Toronto Mentorship Program, Department of Mathematics, University of Toronto

References

References are available upon request.